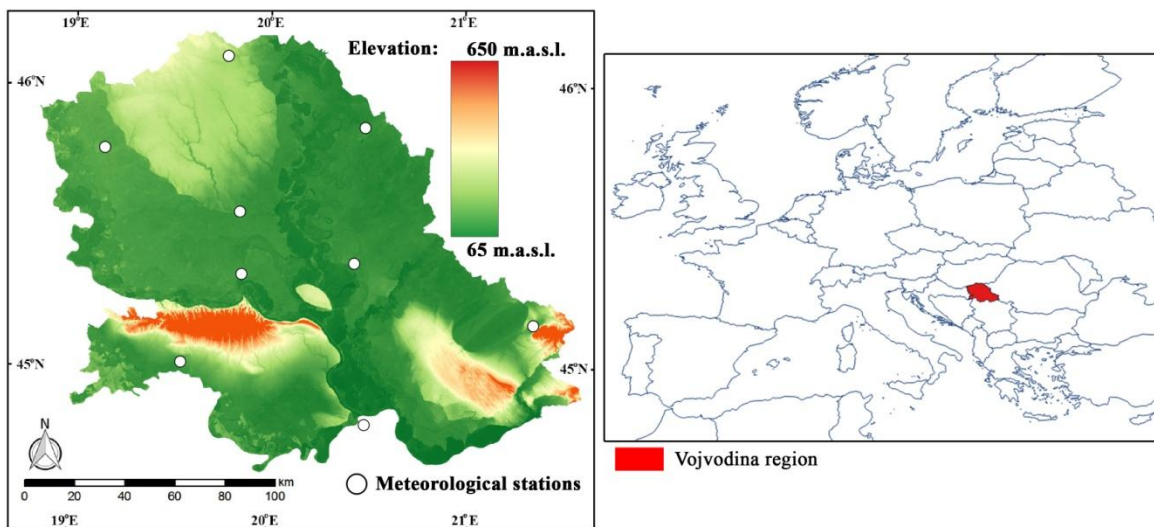


Suše u Vojvodini: edukativni vodič kroz osnovne pojmove vezane za sušu

Region Vojvodine

Region Vojvodine u Srbiji, nalazi se u jugoistočnoj Evropi i zauzima površinu od približno 21.500 km². Region, smešten između 44,6°N i 46,2°N geografske širine i 18,9°E i 21,5°E geografske dužine, odlikuje se umereno-kontinentalnom klimom sa hladnim zimama, toplim i vlažnim letima, izraženim sezonskim kolebanjima temperature i neravnomernom raspodelom padavina. Vojvodina ima blagu umerenu/mezotermalnu klimu sa značajnim padavinama tokom cele godine, suvim zimama i sekundarnim maksimumom padavina u jesen (Bezdan et al., 2024).



Topografska karta regiona Vojvodine

Prostorna i vremenska raspodela padavina u regionu je neravnomerna. Prosečna godišnja količina padavina iznosi 615 mm, od čega 249 mm tokom vegetacionog perioda i 366 mm tokom vanvegetacionog perioda. Srednja godišnja temperatura vazduha iznosi 11,5 °C, povećava se na 18,3 °C tokom vegetacionog perioda i opada na 4,7 °C tokom vanvegetacionog perioda.

Pregled razvoja suše i njenih uticaja u Srbiji

Tokom poslednjih nekoliko decenija Srbija se suočava sa povećanom učestalošću i intenzitetom suša. Ovaj trend je naročito izražen nakon 2000. godine. Iako je suša deo prirodnog klimatskog ciklusa, ovaj nepovoljan trend može se povezati sa signalom klimatskih promena koji se ogleda u pozitivnom trendu temperature i promenama u preraspodeli padavina unutar godišnjeg ciklusa. Većina suša bila je izazvana nedostatkom padavina tokom toplijeg dela godine, praćenim povećanom evapotranspiracijom usled viših

temperatura. Preliminarna procena kumulativnih gubitaka koji se mogu povezati sa sušama nakon 2000. godine premašuje 5 milijardi evra (Drought Initiative - Republic of Serbia, 2020).

Suša je složen fenomen koji može imati brojne negativne uticaje na prirodu i društvo, naročito kada su sušni uslovi neuobičajeno dugotrajni. U slučaju ekstremnih suša, signal može biti toliko snažan da izazove društvene i prirodne nestabilnosti velikih razmera. Prema Svetskoj meteorološkoj organizaciji, „suša je produženi sušni period u prirodnom klimatskom ciklusu koji se može javiti bilo gde u svetu. To je pojava koja se sporo razvija i nastaje usled nedostatka padavina“.

Iako je suša deo prirodne varijabilnosti klimatskog sistema, usled klimatskih promena pojedini delovi sveta već se suočavaju sa povećanom učestalošću i intenzitetom suša, a prema projekcijama klimatskih promena ovaj signal će u budućnosti postajati sve izraženiji i potencijalno će se pojavljivati na većem broju područja. Ovakva situacija predstavlja poziv za bolje strateško planiranje u pogledu smanjenja rizika od suše i prilagođavanja takvim uslovima.

U Srbiji je suša u prošlosti zabeležena mnogo puta, ali se od početka 21. veka zemlja suočila sa nekoliko snažnih epizoda suše koje su prouzrokovale velike ekonomske gubitke. Od 2000. godine ukupni gubici i štete izazvani ekstremnim vremenskim i klimatskim događajima premašuju 5 milijardi evra, a više od 70% tih gubitaka povezano je sa sušom i visokim temperaturama (Drought Initiative - Republic of Serbia, 2020).

Razumevanje suše: meteorološka, poljoprivredna, hidrološka i socioekonomska suša

Meteorološka suša se definiše kao nedostatak padavina, odnosno, suša se definiše kao period neuobičajeno suvog vremena koji traje dovoljno dugo da izazove ozbiljnu hidrološku neravnotežu. Iz ove definicije jasno je da su potrebne referentne vrednosti padavina iz prethodnih osmatranja kako bi se procenilo da li se trenutno stanje može kategorisati kao neuobičajeno. Definicija takođe ukazuje na to da ovaj neuobičajeno sušan period mora trajati dovoljno dugo da utiče na druge komponente klimatskog sistema, odnosno da u ovom slučaju izazove hidrološku neravnotežu.

Pošto meteorološka suša zavisi od određenih referentnih, očekivanih vrednosti, ona je regionalno i vremenski zavisna, jer se u različitim delovima sveta i u različitim delovima godine očekuju različite količine padavina. To je verovatno jedan od razloga zbog kojih je Standardizovani indeks padavina (SPI), koji meri odstupanje od očekivanih padavina u određenom regionu i delu godine, široko korišćen za karakterizaciju meteorološke suše na različitim vremenskim skalama. SPI omogućava poređenje uslova suše između regiona sa izrazito različitim klimama. On kvantifikuje osmotrene padavine kao standardizovano odstupanje od izabrane funkcije raspodele verovatnoće koja modelira osmotrene podatke o padavinama. SPI se može tumačiti kao broj standardnih devijacija za koji osmotrena anomalija odstupa od dugoročnog proseka. SPI se često izračunava za periode od 1 do 36 meseci. U operativnoj praksi SPI je prepoznat kao standardni indeks koji bi trebalo da bude dostupan širom sveta za kvantifikovanje i izveštavanje o meteorološkoj suši. Ipak, izražena je zabrinutost u vezi sa primenom SPI-ja kao mere

promena suše povezanih sa klimatskim promenama, jer ne uzima u obzir promene evapotranspiracije. Zbog toga su predloženi alternativni indeksi koji uključuju evapotranspiraciju (Drought Initiative - Republic of Serbia, 2020).

Poljoprivredna suša nastaje kada različite karakteristike meteorološke suše, proistekle iz nedostatka padavina, u kombinaciji sa drugim potencijalno negativnim uticajima, kao što su razlike između stvarne i potencijalne evapotranspiracije, deficit vode u zemljištu, sniženi nivoi podzemnih voda ili akumulacija i dr., izazovu posledice u poljoprivredi. U tom smislu, poljoprivredna suša je složeniji fenomen od meteorološke suše, jer u slučajevima kada se deficit padavina karakteriše kao sušan, ali ne i ekstremno sušan, drugi faktori, poput prekomerne evapotranspiracije usled visokih temperatura, mogu dovesti do dodatnog stresa biljaka. Pored toga, biološke karakteristike određene biljke i poklapanje suvih meteoroloških uslova sa fenološkim fazama razvoja biljaka mogu dovesti do različitih posledica. Deficit vlage u površinskom sloju zemljišta u vreme setve može ugroziti klijanje. S druge strane, potencijalne negativne anomalije vlage u dubljim slojevima zemljišta u tim ranim fazama neće uticati na konačan prinos ako se vlaga u dubljim slojevima obnovi u odgovarajućem trenutku tokom kasnijih faza razvoja. Precizno određivanje poljoprivredne suše trebalo bi da uzme u obzir sve ove različite aspekte (Drought Initiative - Republic of Serbia, 2020).

Hidrološka suša predstavlja stanje u kojem nedostatak padavina, uključujući smanjenu akumulaciju snega i snežni pokrivač, dovodi do smanjenja površinskih i podzemnih vodnih resursa. Hidrološka suša se često proučava i prati na nivou rečnog sliva. Pošto je potrebno određeno vreme da deficit padavina utiče na komponente hidrološkog sistema, kao što su vlažnost zemljišta, rečni proticaji ili nivoi podzemnih voda i akumulacija, postoji vremensko kašnjenje između pojave meteorološke i poljoprivredne suše i hidrološke suše. Negativna anomalija padavina može relativno brzo dovesti do smanjenja vlažnosti zemljišta, što lako može uticati na biljke i izazvati poljoprivrednu sušu, ali je potrebno više vremena da se posledice ovog deficita uoče u nivoima podzemnih voda ili proticajima velikih reka. Kod hidroloških suša dodatne komplikacije proizlaze iz činjenice da se vodni resursi često koriste za višestruke i međusobno konkurentne namene (npr. navodnjavanje, hidroenergetiku, vodosnabdevanje stanovništva i industrije itd.), pa je ponekad teško razlikovati i kvantifikovati uticaje prirodnih i antropogenih izvora. Tokom perioda suše može doći do konkurencije za vodu (Drought Initiative - Republic of Serbia, 2020).

Socioekonomska suša nastaje kada kombinovani negativni uticaji prethodno opisana tri tipa suše deluju istovremeno, dovodeći do problema u ponudi i potražnji za određenim ekonomskim dobrom. Ona se razlikuje od prethodno navedenih tipova suše jer njena pojava zavisi od vremenskih i prostornih procesa ponude i potražnje. Ponuda mnogih ekonomskih dobara, kao što su voda, stočna hrana, žitarice, riba i hidroelektrična energija, zavisi od vremenskih uslova. Zbog prirodne varijabilnosti klime, u nekim godinama raspoloživost vode je velika, dok u drugim nije dovoljna da zadovolji potrebe ljudi i životne sredine. Socioekonomska suša nastaje kada potražnja za nekim ekonomskim dobrom premaši ponudu kao posledica vremenski uslovljenog nedostatka vode (<http://drought.unl.edu/Education/DroughtIn-depth/TypesofDrought>). Jasno je da vremenski sled razvoja različitih tipova suše vodi od deficita padavina i razvoja meteorološke suše, preko poljoprivredne i hidrološke suše, do uslova koji se mogu okarakterisati kao socioekonomska suša. Ovakav scenario bio je evidentan tokom suše u Srbiji 2012. godine, kada su bila prisutna sva četiri tipa suše.

Na primer, deficit padavina može dovesti do meteorološke suše, dok povećana evapotranspiracija usled visoke temperature može dovesti do poljoprivredne suše, na koju mogu uticati i tip zemljišta ili vrsta vegetacije na posmatranom području. Različiti tipovi zemljišta i različita vegetacija (usevi) mogu različito reagovati na meteorološku sušu (Cook et al., 2018; Van Loon, 2015).

Indeksi i indikatori suše

Ne postoji jedna metrika ili jedna promenljiva kojom se suša može meriti/pratiti, već se ona procenjuje pomoću izvedenih pokazatelja, takozvanih indeksa. Zbog veoma složene prirode suše, slično kao i kod njenog definisanja, postoji veliki broj indeksa koji se koriste za praćenje suše.

U nastavku će biti prikazano nekoliko indeksa koji se već koriste u Srbiji u operativne ili istraživačke svrhe. Indeksi koji se koriste za praćenje suše u Srbiji su Standardizovani indeks padavina (SPI), Standardizovani indeks padavina i evapotranspiracije (SPEI), Palmerov indeks ozbiljnosti suše (PDSI) i Palmerov Z indeks. Kako bi se prikazali različiti pristupi karakterizaciji različitih tipova suše, biće predstavljeni i dodatni indeksi i indikatori koji se često koriste.

Standardizovani indeks padavina (SPI). Svetska meteorološka organizacija (WMO) je preporučila SPI kao glavni indeks meteorološke suše koji zemlje treba da koriste za praćenje uslova suše. Prepoznavanjem SPI-ja kao indeksa za široku primenu, WMO je pružila smernice zemljama koje nastoje da uspostave sistem ranog upozoravanja na sušu. Indeks uključuje istorijske zapise o padavinama za bilo koju lokaciju radi određivanja verovatnoće padavina i može se izračunavati na različitim vremenskim skalama, od 1 meseca do 48 meseci ili duže.

Standardizovani indeks padavina i evapotranspiracije (SPEI). Kao relativno nov indeks suše, SPEI se zasniva na SPI-ju, ali uključuje i temperaturnu komponentu, čime omogućava da se kroz osnovni proračun vodnog bilansa uzme u obzir uticaj temperature na razvoj suše. SPEI ima skalu intenziteta na kojoj se izračunavaju i pozitivne i negativne vrednosti, čime se identifikuju vlažni i sušni događaji. Može se izračunavati za vremenske korake od samo 1 meseca do 48 meseci ili duže. Mesečno ažuriranje omogućava njegovu operativnu primenu, a što je vremenska serija raspoloživih podataka duža, rezultati su pouzdaniji. Uz istu fleksibilnost kao SPI, SPEI se može koristiti za identifikovanje i praćenje uslova povezanih sa različitim posledicama suše. Uključivanje temperature zajedno sa podacima o padavinama omogućava SPEI-ju da uzme u obzir uticaj temperature na stanje suše, što može biti posebno važno jer se tokom poslednjih nekoliko decenija, usled klimatskih promena, beleži dugoročni pozitivan trend temperature.

Indeks uzastopnih sušnih dana (CDD) definiše se kao maksimalno trajanje sušnog perioda, izraženo najvećim brojem uzastopnih dana sa dnevnim padavinama manjim od 1 mm tokom određenog perioda (npr. sezone ili godine). Indeks se često koristi za karakterizaciju uslova meteorološke suše i procenu promena učestalosti, trajanja i intenziteta suše u osmotrenim i projektovanim klimatskim uslovima.

Palmerov indeks ozbiljnosti suše (PDSI). Izračunava se korišćenjem mesečnih podataka o temperaturi i padavinama, zajedno sa informacijama o kapacitetu zemljišta za zadržavanje vode. U obzir uzima

primljenu vlagu (padavine), kao i vlagu uskladištenu u zemljištu, uzimajući u obzir potencijalni gubitak vlage usled uticaja temperature. Razvijen je prvenstveno za identifikovanje suša koje utiču na poljoprivredu.

Palmerov Z indeks. Palmerov Z indeks bolje reaguje na kratkoročne uslove od PDSI-ja i obično se izračunava za kraće vremenske skale, što mu omogućava da identifikuje suše koje se brzo razvijaju.

Indeks vlažnosti useva (CMI). Kao deo originalnog Palmerovog rada iz ranih šezdesetih godina, CMI se obično izračunava nedeljno zajedno sa Palmerovim indeksom ozbiljnosti suše (PDSI), kao kratkoročna komponenta suše u kojoj se razmatra uticaj na poljoprivredu. Koristi se za praćenje suša kod kojih su uticaji na poljoprivredu od primarnog značaja.

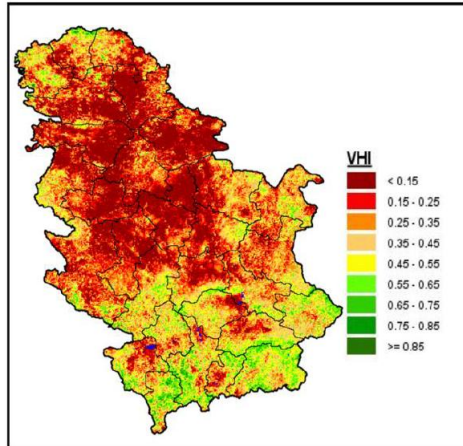
Anomalija vlažnosti zemljišta (SMA). Razvili su je Bergman i saradnici u Nacionalnoj meteorološkoj službi Sjedinjenih Američkih Država sredinom osamdesetih godina kao način procene globalnih uslova suše. Može koristiti nedeljne ili mesečne vrednosti padavina i potencijalne evapotranspiracije u jednostavnoj jednačini vodnog bilansa. Namenjena je da prikaže stepen isušenosti ili zasićenosti zemljišta u poređenju sa normalnim uslovima i da pokaže kako stres usled nedostatka vlage u zemljištu utiče na proizvodnju useva širom sveta. Razvijena je i široko se koristi za praćenje uticaja suše na poljoprivredu i biljnu proizvodnju širom sveta.

Indeks suše proticaja (SDI). Razvili su ga Nalbantis i Tsakiris (2009), koristeći metodologiju i proračune SPI-ja kao osnovu. Koristi mesečne vrednosti proticaja i metode normalizacije povezane sa SPI-jem za razvoj indeksa suše zasnovanog na podacima o proticaju. Sa rezultatima sličnim SPI-ju, mogu se analizirati i vlažni i sušni periodi, kao i intenzitet tih pojava. Koristi se za praćenje i identifikovanje događaja suše u odnosu na određenu mernu stanicu, koja može, ali ne mora, predstavljati veći sliv. Može se koristiti za praćenje hidrološke suše.

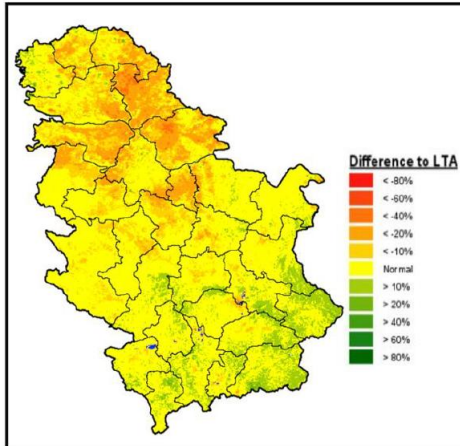
Normalizovani indeks razlike vegetacije (NDVI). Za izračunavanje NDVI-ja koriste se globalni podaci o indeksu vegetacije, dobijeni mapiranjem dnevne radjanse rezolucije 4 km. Vrednosti radjanse izmerene u vidljivom i bliskom infracrvenom delu spektra koriste se za izračunavanje NDVI-ja. Indeks meri zelenilo i vitalnost vegetacije tokom sedmodnevnog perioda, čime se smanjuje uticaj oblačnosti, i može identifikovati stres vegetacije povezan sa sušom. Koristi se za identifikovanje i praćenje suša koje utiču na poljoprivredu.

Indeks zdravlja vegetacije (VHI). Jedan od prvih pokušaja praćenja i identifikovanja uticaja suše na poljoprivredu korišćenjem podataka daljinske detekcije. Podaci sa radiometra vrlo visoke rezolucije (AVHRR) u vidljivom, infracrvenom i bliskom infracrvenom delu spektra koriste se za identifikovanje i klasifikovanje stresa vegetacije izazvanog sušom. Koristi se za identifikovanje i praćenje suša koje utiču na poljoprivredu širom sveta. Primer VHI-ja i anomalije NDVI-ja za Srbiju tokom letnje suše 2017. godine prikazan je na slici ispod.

Vegetation Health Index



NDVI Anomaly



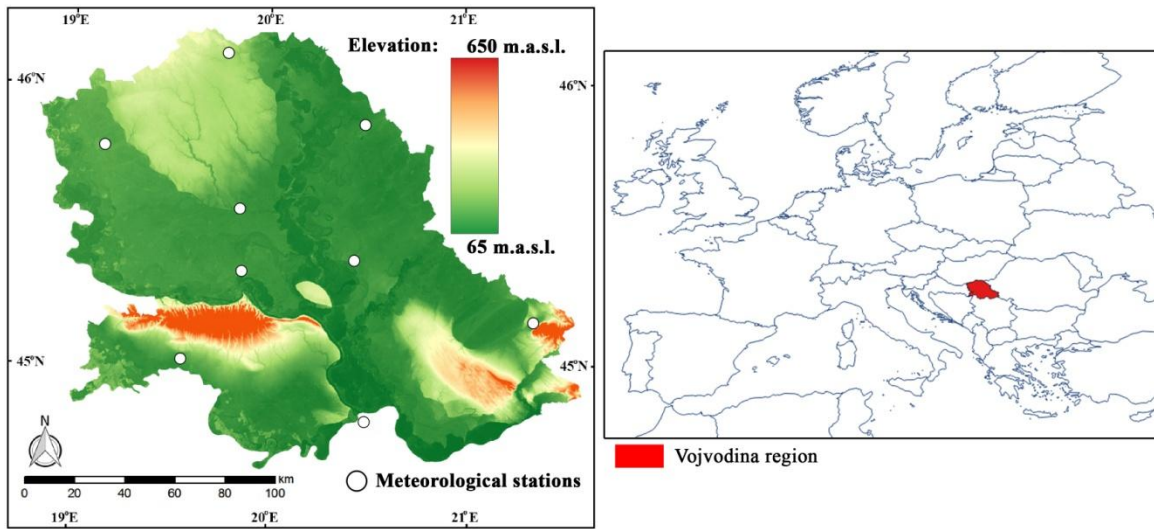
Indeks zdravlja vegetacije (levo) i anomalija NDVI-ja (desno) za Srbiju, za „drugu dekadu“ avgusta 2017. godine, koji ukazuju na snažnu poljoprivrednu sušu. Izvor: FAO

Postojeći glavni sistem za praćenje suše u Srbiji, u okviru Republičkog hidrometeorološkog zavoda Srbije, zasniva se na Standardizovanom indeksu padavina (SPI). Dodatni indeksi suše koji se izračunavaju za druge biltene ove institucije su Palmerov indeks ozbiljnosti suše (PDSI), Palmerov Z indeks i sadržaj vlage u zemljištu (Drought Initiative - Republic of Serbia, 2020).

Droughts in Vojvodina: An Educational Guide to Basic Drought-Related Concepts

Vojvodina Region

Vojvodina region in Serbia situated in southeast Europe and covering approximately 21,500 km². The region, located between 44.6°N to 46.2°N latitude and 18.9°E to 21.5°E longitude, experiences a temperate continental climate with cold winters, warm and humid summers, significant seasonal temperature fluctuations, and uneven precipitation distribution. According to the Köppen climate classification, the Vojvodina region's climatological formula is Cfbw^x", indicating mild temperate/mesothermal climate with significant year-round precipitation, dry winters, and a secondary precipitation peak in autumn (Bezdan et al., 2024).



Topographic map of the Vojvodina region

Spatial and temporal precipitation distribution in the region is uneven. The average annual precipitation is 615 mm, divided into 249 mm during the growing season and 366 mm in the non-growing season. The mean annual air temperature is 11.5 °C, rising to 18.3 °C in the growing season and dropping to 4.7 °C in the non-growing season.

Overview of Drought Development and Impacts in Serbia

Over the last several decades Serbia faces increased frequency of drought and drought severity. This trend is especially noticeable after 2000. Even that drought is a part of natural climate cycle, this unfavorable trend can be related to the climate change signal of positive temperature trend and changes in redistribution of precipitation within annual cycle. Most of the droughts were triggered by lack of the precipitation during warmer part of the year, accompanied with increased evapotranspiration due to

increased temperature. Preliminary estimate of cumulative losses that can be related to the droughts after 2000, exceeds 5 billion euros (Drought Initiative - Republic of Serbia, 2020).

Drought is a complex phenomenon that can have many negative impacts on nature and society, especially when dry conditions are unusually long. In case of extreme droughts, the signal can be so strong to trigger social and nature instabilities on a large scale. According to World Meteorological Organization “drought is a prolonged dry period in the natural climate cycle that can occur anywhere in the world. It is a slow on-set phenomenon caused by a lack of rainfall” (<https://public.wmo.int/en/our-mandate/water/drought>).

Even that drought is a part of natural variability within the climate system, due to the climate change some parts of the world already faced increased frequency and intensity of droughts, and according to climate change projections this signal will become stronger in the future, and potentially will emerge on more places. This situation can be seen as a call for better strategic planning in terms of drought risk reduction and adaptation on such conditions.

In Serbia drought was observed many times in past, but since beginning 21st century Serbia faced several strong drought episodes that induced/caused great economic losses. According to the Serbia's Intended Nationally Determined Contribution (INDC), submitted before adoption of Paris agreement, since 2000 total loss and damage caused by extreme weather and climate events exceeds 5 billion euros, and more than 70% of these losses are associated with drought and high temperatures (Drought Initiative - Republic of Serbia, 2020).

Understanding Drought: Meteorological, Agricultural, Hydrological and Socio-economic Drought

In the early 1980s more than 150 published definitions of drought was found (<http://drought.unl.edu>). Different definitions were driven by differences related to the regions, needs, and disciplinary approaches. Analyzing different definitions of drought Wilhite and Glantz (1985) categorize them in four basic approaches: meteorological, agricultural, hydrological, and socioeconomic. The meteorological, agricultural and hydrological, try to detect drought as a geophysical phenomenon, and the socioeconomic tries to define drought in terms of supply and demand, tracking the effects of water shortfall as it ripples through socioeconomic systems.

Meteorological drought is virtually always defined as a lack of precipitation. According to AMS Glossary of Meteorology drought is defined as period of abnormally dry weather sufficiently long enough to cause a serious hydrological imbalance. From this definition is it clear that some reference precipitation values from past observation is needed as a measure that current condition can be categorized as abnormal. Also, definition lead us to the fact that for drought this unusual dry period should be long enough to alter other components of the climate system, in this case to induce hydrological imbalance.

Since that meteorological drought depend on some reference, expected values, it is regional and time dependent, because in the different part of the world and in the different part of the year, different amount of precipitation is expected. This situation probably leads to the situation that the Standardized

Precipitation Index (SPI), which measures departure from expected precipitation in some region and part of the year is a widely used index to characterize meteorological drought on a different range of timescales. SPI allows to compare drought condition across regions with markedly different climates. It quantifies observed precipitation as a standardized departure from a selected probability distribution function that models the observed precipitation data. SPI can be interpreted as the number of standard deviations by which the observed anomaly deviates from the long-term mean. SPI is often calculated for differing periods from 1 up to 36 months. For the operational community, the SPI has been recognized as the standard index that should be available worldwide for quantifying and reporting meteorological drought. Concerns have been raised about the utility of the SPI as a measure of changes in drought associated with climate change, as it does not deal with changes in evapotranspiration. Because of this alternative indices that deal with evapotranspiration have been proposed (Drought Initiative - Republic of Serbia, 2020).

Agricultural drought is triggered when various characteristics of meteorological drought, that come from lack of precipitation, in combination with other potential negative impacts such as differences between actual and potential evapotranspiration, soil water deficits, reduced groundwater or reservoir levels, etc. create impact in agricultural. In that sense, agricultural drought is more complex phenomenon in comparison to meteorological drought, because in cases when precipitation deficits are characterized as dry but not extreme dry, other factors such as excessive evapotranspiration, due to high temperature, can lead to additional stress on plants. In addition, biological characteristics of the specific plant, and coincidence of dry meteorological condition with phenological stages of plant development can lead to different impacts. Top soil moisture deficit at planting can jeopardize germination. On the other hand, potential negative anomalies in subsoil moisture at these early stages will not affect final yield if subsoil moisture is replenished at a right time during later development stages. A precise determination of agricultural drought should be able to account these different aspects (Drought Initiative - Republic of Serbia, 2020).

Hydrological drought is situation when precipitation shortages, including reduction in snow accumulation and snow pack, results in decrease in surface and subsurface water resources. Hydrological droughts are often studied and monitored on a river basin scale. Since it takes some time that precipitation deficiencies make some impact on components of the hydrological system such as soil moisture, river discharges, or groundwater and reservoir levels, there is some time lag in onset of meteorological and agricultural droughts and hydrological droughts. A negative anomaly in precipitation may lead relatively fast to soil moisture decrease that can easily impact plants, and onset agricultural drought, but it will take some time to see the impact of this deficiency in groundwater levels, or discharges on large rivers. In case of hydrological droughts, additional complications can come from the fact that water resources are often used for multiple and competing purposes (e.g., irrigation, hydropower, domestic and industrial water use etc.) and sometimes is hard to distinguish and quantify impacts from natural and anthropogenic sources. During the periods of drought competition for water may occur (Drought Initiative - Republic of Serbia, 2020).

Socioeconomic drought occurs when combined negative impacts of previously described three types of droughts act in the same time, leading to problems of supply and demand of some economic good. It differs from the aforementioned types of drought because its occurrence depends on the time and space

processes of supply and demand. The supply of many economic goods, such as water, forage, food grains, fish, and hydroelectric power, depends on weather. Because of the natural variability of climate, water supply is ample in some years but unable to meet human and environmental needs in other years. Socioeconomic drought occurs when the demand for an economic good exceeds supply as a result of a weather-related shortfall in water supply (<http://drought.unl.edu/Education/DroughtIn-depth/TypesofDrought>). It is clear that time line of the developing of different types of drought leads from precipitation deficit and development of meteorological drought over agricultural and hydrological drought to the conditions that can be characterized as a socioeconomic drought (Figure 5). This kind of scenario is evident for the drought in Serbia in 2012, when all four types of drought were present.

For example, deficit in precipitation can lead to meteorological drought, but increased evapotranspiration due to high temperature can lead to agricultural drought, which can be also affected by soil type, or type of vegetation over area of interest. Different soil types and different vegetation (crops) can have different response to meteorological drought (Cook et al., 2018; Van Loon, 2015).

Drought Indices and indicators

There is no single metric or single variable that can be used to measure/monitor the drought, but it is measured by derived metrics, so-called indices. Because of very complex nature of drought, similar to the definition of it, there is almost endless number of indices that are used for drought monitoring. Also, many indices are developed to monitor different type of droughts. Extensive list of different indices that can be used for drought monitoring can be found on the web site of the World Meteorological Organization and Global Water Partnership - Integrated Drought Management Programme (<http://www.droughtmanagement.info/indices>).

Listed indices can be sorted following type of drought that can be monitored using selected index, easiness of use, and required input variables/parameters that are needed for calculation. Also for every index there is short explanation of origin, basic characteristics, description of applications, strengths and weaknesses, list of the countries where index is in the use, appropriate scientific reference, and very important information link to the data base or resource from which index can be use.

Here will be reviewed several of them that are already used in Serbia for operational or research purposes. Indices that are in use for drought monitoring in Serbia are Standardized Precipitation Index (SPI), Standardized Precipitation Evapotranspiration Index (SPEI), Palmer Drought Severity Index (PDSI), Palmer Z Index. To demonstrate different approaches for characterization of different drought types additional indices and indicators that are commonly in use will be also presented.

Standardized Precipitation Index (SPI). WMO recommended SPI as the main meteorological drought index that countries should use to monitor and follow drought conditions. By identifying SPI as an index for broad use, WMO provided direction for countries trying to establish a level of drought early warning. Index includes historical precipitation records for any location to develop a probability of precipitation and can be computed at any number of timescales, from 1 month to 48 months or longer. Drought events are indicated when the results of SPI, for whichever timescale is being investigated, become continuously

negative and reach a value less than or equal to -1. The drought event is considered to be ongoing until SPI reaches a value of 0. SPI is used by Republic Hydrometeorological Service in Serbia for drought monitoring in country.

Standardized Precipitation Evapotranspiration Index (SPEI). As a relatively new drought index, SPEI uses the basis of SPI but includes a temperature component, allowing the index to account for the effect of temperature on drought development through a basic water balance calculation. SPEI has an intensity scale in which both positive and negative values are calculated, identifying wet and dry events. It can be calculated for time steps of as little as 1 month up to 48 months or more. Monthly updates allow it to be used operationally, and the longer the time series of data available, the more robust the results will be. With the same versatility as that of SPI, SPEI can be used to identify and monitor conditions associated with a variety of drought impacts. The inclusion of temperature along with precipitation data allows SPEI to account for the impact of temperature on a drought situation, which can be especially important, since that during last several decades long term positive temperature trend is observed due to climate change.

The Consecutive Dry Days (CDD) index is defined as the maximum length of a dry spell, expressed as the highest number of consecutive days with daily precipitation below 1 mm during a given period (e.g., a season or year). The index is commonly used to characterize meteorological drought conditions and to evaluate changes in drought frequency, duration, and severity under observed and projected climate conditions.

Palmer Drought Severity Index (PDSI). Calculated using monthly temperature and precipitation data along with information on the water-holding capacity of soils. It takes into account moisture received (precipitation) as well as moisture stored in the soil, accounting for the potential loss of moisture due to temperature influences. Developed mainly as a way to identify droughts affecting agriculture, but it has also been used for identifying and monitoring droughts associated with other types of impacts. With the longevity of PDSI, there are numerous examples of its use over the years.

Palmer Z Index The Palmer Z Index responds to short-term conditions better than PDSI and is typically calculated for shorter timescales, enabling it to identify rapidly developing drought conditions. As part of the original work done by Palmer in the early 1960s, the Palmer Z Index is usually calculated on a monthly basis along with PDSI output as the moisture anomaly. The Palmer Z Index is a derivative of PDSI and the Z values are part of the output.

Crop Moisture Index (CMI) As part of original work done by Palmer in the early 1960s, CMI is usually calculated weekly along with the Palmer Drought Severity Index (PDSI) output as the short-term drought component in which the impact on agriculture is considered. Used to monitor droughts in which agricultural impacts are a primary concern.

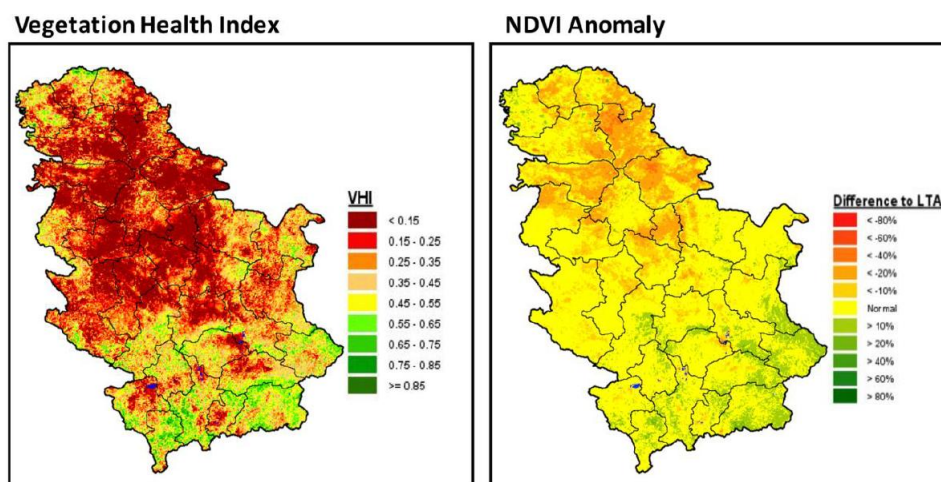
Soil Moisture Anomaly (SMA) Developed by Bergman et al. at the National Weather Service in the United States during the mid-1980s as a way to assess global drought conditions. Can use weekly or monthly precipitation and potential evapotranspiration values in a simple water balance equation. It is intended to reflect the degree of dryness or saturation of the soil compared with normal conditions and to

show how soil moisture stress influences crop production around the world. Developed and used extensively for monitoring drought impacts on agriculture and crop production around the world.

Streamflow Drought Index (SDI). Developed by Nalbantis and Tsakiris (2009) using the methodology and calculations of SPI as the basis for their work. Uses monthly streamflow values and the methods of normalization associated with SPI for developing a drought index based upon streamflow data. With an output similar to that of SPI, both wet and dry periods can be investigated, as well as the severity of these occurrences. Used to monitor and identify drought events with reference to a particular gauge, which may or may not represent larger basins. Can be used to monitor hydrological drought.

Normalized Difference Vegetation Index (NDVI). For NDVI calculation the global vegetation index data is used, which are produced by mapping 4 km daily radiance. Radiance values measured in both the visible and near-infrared channels are used to calculate NDVI. It measures greenness and vigour of vegetation over a seven-day period as a way of reducing cloud contamination and can identify drought-related stress to vegetation. Used for identifying and monitoring droughts affecting agriculture.

Vegetation Health Index (VHI). One of the first attempts to monitor and identify drought-related agricultural impacts using remotely sensed data. Advanced Very High Resolution Radiometer (AVHRR) data in the visible, infrared and near-infrared channels are all used to identify and classify stress to vegetation due to drought. Used to identify and monitor droughts affecting agriculture around the world. An example of VHI and NDVI Anomaly for Serbia, during summer drought in 2017 is presented on figure below.



Vegetation health index (left) and NDVI anomaly (right), over Serbia, for 'dekad 2' in August 2017, indicating strong agricultural drought. Source: FAO

Existing main drought monitoring system in Serbia, within Republic Hydrometeorological Service of Serbia, is based on Standardized Precipitation Index (SPI), additional drought indices that are calculated for other bulletins published by this institution, are Palmer Drought Severity Index (PDSI), Palmer Z index and soil moisture content (Drought Initiative - Republic of Serbia, 2020).

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